

Melika Shahghadami

Technical Artist | 3D Artist | Unity | VR | Unreal Engine

Technical Art • 3D Modeling • Unity & Unreal Engine • VR • Optimization

Unity & Unreal Engine

Technical Art & Optimization

VR • Meta Quest 3



Skills

3D Art

High/Low Poly Modeling • Retopology • UV Mapping • PBR Texturing • Baking • Materials • Mesh Optimization

Technical Art

Unity URP • Shader Graph • VFX / Particle Systems • Scene & Lighting Setup • Profiling & Performance Optimization • Editor Tools

Character Art

Rigging • Skinning • Animation Controllers

VR

Meta Quest 3 Optimization • Real-time Performance

Software

Primary

Blender

Unity

Unreal Engine

Substance Painter

Additional

Photoshop

Maya

Character Creator

iClone

PureRef

Illustrator

Key Projects

Industrial Warehouse — Unity URP

PBR Materials • Volumetric Lighting • Environment Art

VR Driving Simulator — Unity / Meta Quest 3

Day/Night Cycle • Environment Art • Optimization

Multiplayer Shooter — Unity / VR / FPS

Level Design • Cover Placement • Sightlines

Cinematic Environment — Blender / Cycles

PBR • Volumetric Fog • Particle Systems

About Me

Technical Artist and 3D Artist with 3+ years of experience producing and preparing real-time assets for games, simulators and VR projects. Experienced in modeling and optimization, PBR texturing, rigging, and technical asset implementation in Unity and Unreal Engine. Focused on balancing visual quality and performance, especially in VR projects and on Meta Quest 3.

Work Experience

Ario Sorun — 3D Designer & Technical Artist

2023 – Present

- Asset implementation in Unity, scene layout, material setup and lighting.
- Mesh, texture and draw call optimization to improve performance in VR projects on Meta Quest 3.
- Developing and standardizing the asset production pipeline for better coordination between Art and Development teams.

Pishgaman Lotus — 3D Designer & Godot Programmer

2022 – 2023

- Environment and modular 3D asset design for game projects.
- Collaborated with the development team on technical implementation and final build preparation.

Tarsim Dadeh — Java Developer

2021 – 2022

- Developed and maintained software modules in Java; debugging and code optimization.

Education

- B.Sc. Computer Engineering — Shahid Beheshti University (2016–2021)
- M.Sc. IT Management — University of Tehran, Markazi Branch (2021–2023)

Industrial Environment in Unity

Industrial Warehouse • Unity • URP

Melika Shahghadami | Technical Artist | Environment Art + Volumetric Lighting



Main work — industrial warehouse environment built in Unity URP

About the Project

Industrial warehouse environment built in Unity using URP, focusing on PBR materials, volumetric lighting, environment art, and the design and layout of industrial props. The scene is optimized for real-time rendering and use in game projects.

Unity URP Environment Lighting PBR Level Art

Technical Specs

URP

Render Pipeline

Volumetric

Volumetric light & fog

Game Ready

Optimized for games

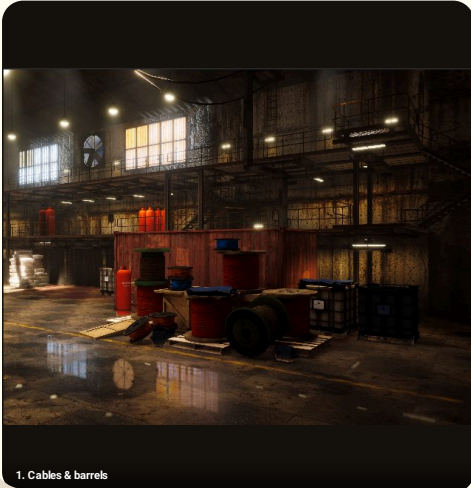
Lighting & Atmosphere

- Volumetric light through glass windows
- Multiple hanging industrial lamps
- Cinematic Post-Processing
- Floor reflections with SSR

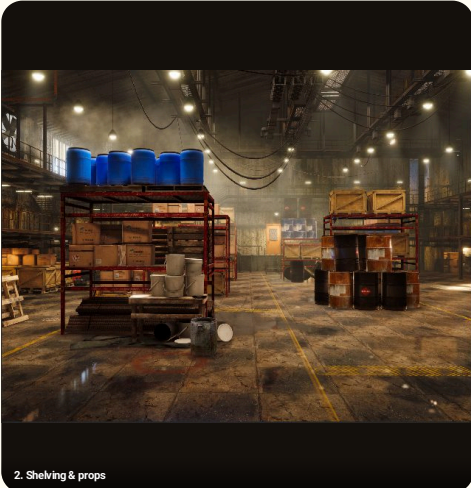
Surfaces & Details

- Industrial shelving and layout
- Overhead crane and steel structure
- Aged, rusty PBR materials
- Varied props: barrels, crates, pallets

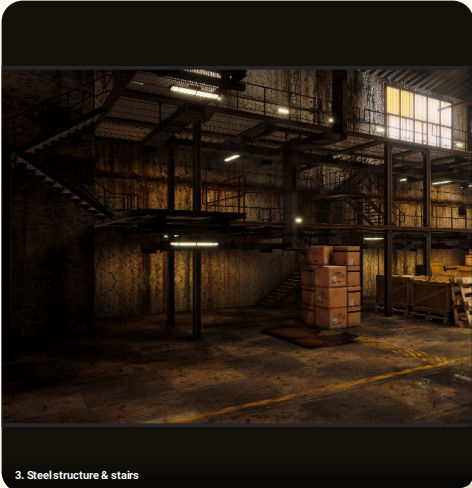
Environment details — 3 warehouse angles — Unity URP



1. Cables & barrels



2. Shelving & props



3. Steel structure & stairs

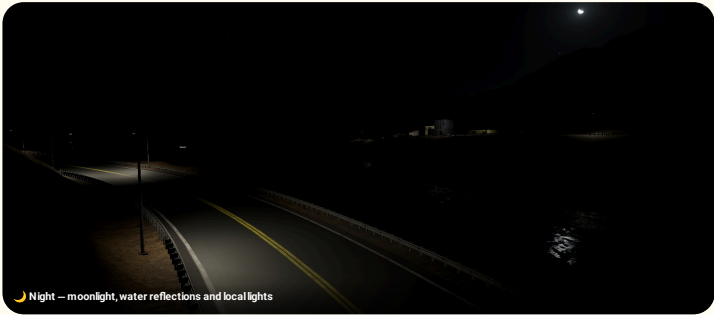
Multiplayer VR Driving Simulator — Day/Night Cycle

Multiplayer VR Driving Simulator • Day / Night Cycle • Meta Quest 3

Melika Shahghadami | Technical Artist | Large-scale multiplayer environment with dynamic lighting



☀ Day — mountain lake with natural light and ambient detail



🌙 Night — moonlight, water reflections and local lights

About the Multiplayer Project

Multiplayer VR driving simulator for Meta Quest 3 with a shared environment and day/night cycle. The environment includes a mountain road, tunnel, lake, vegetation and structures, with dynamic lighting and real-time optimization targeting 72+ FPS.

Multiplayer VR 72+ FPS Day/Night Network Ready Environment Art Lighting Unity Quest 3

Multiplayer Features

- Shared multi-user environment with synchronization
- Draw call and LOD optimization for multiplayer
- Route and environment design for user interaction

Technical Achievements

- Large mountain environment with lake and tunnel
- Day/night lighting with moon and artificial lights
- 40% Draw Call Reduction — Tested on Meta Quest 3

☀ Day Views — Route Variety



Billboard



Aerial



Mountains



Forest

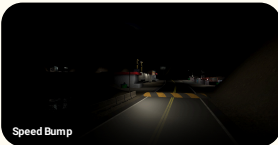


Dirt Road

🌙 Night Views — Lighting System



Tunnel



Speed Bump



Café



Lit Billboard



Building

🔄 Day/Night Comparison — One Location, Two Lightings



Day — Shoreline



Night — Café



Day — Dirt Road



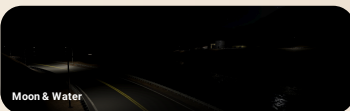
Day — Hill



Panorama



Main Road



Moon & Water



Night Tunnel

Multiplayer Shooter Environment

Multiplayer Shooter Environment • Unity • FPS • Level Design

Melika Shahghadami | Technical Artist | FPS environment design focused on multiplayer gameplay



Overview of the desert environment with adobe architecture and vegetation



Competitive route with cover and sightlines for FPS gameplay

About the Project

First-person multiplayer shooter environment set in a desert with adobe architecture and palm groves. As technical artist, I was responsible for level layout, optimization for multiplayer, and route design with cover and sightlines for balanced gameplay. The environment is designed in Unity for multiplayer FPS, focusing on route readability, player flow and performance.

Multiplayer FPS Level Design Unity Environment Art Optimization

Multiplayer Features

- Route design with cover placement and gameplay-focused sightlines
- Environment optimization for real-time multiplayer
- Sightlines and player flow

Technical Achievements

- Level layout focused on gameplay
- PBR texturing and desert lighting
- LOD and draw call optimization



Cover & Side Route View



Competitive View – Main Route



Desert



Adobe



Palms



FPS Route

Cinematic 3D Environment

Cinematic 3D Environment • Blender • Cycles • Atmospheric Lighting

Melika Shahghadami | Technical Artist | Main piece + lighting variant



Main work — full cinematic environment with windmill, abandoned hut, stone tower and lake — Blender Cycles

About the Project

Personal cinematic environment built in Blender combining a windmill, abandoned hut, stone tower and lake. The scene is created with Cycles, PBR materials, atmospheric lighting, volumetric fog and particle systems, and features water with reflections and displacement.

Blender Cycles PBR Main Lighting Variant Cinematic

Main Scene Features

- 4 elements combined in one cinematic scene
- Grass and reeds via particle systems
- Water with reflections and displacement
- Volumetric fog and atmospheric lighting

Lake Features

- Calm lake with full reflections
- Reeds via particle system
- Water with reflections and displacement
- Overcast daytime mood, no fog

Lighting variant — lake and reeds — Blender / Cycles



Lake and reeds — overcast day, still water

Wolf Character — Modeling & Fur

Wolf Character • Blender Particle Hair • Fur Grooming • PBR Texturing

Melika Shahghadami | Technical Artist | Modeling + Particle Hair / Fur Grooming



Final render — three-quarter view with studio lighting and fur detail



Final render — side view with PBR texturing and fur grooming

About the Project

Wolf character project focused on high/low poly modeling, retopology, UVs, PBR texturing and fur grooming in Blender. The fur is built with particle hair and grooming, and the model is prepared for both real-time and offline rendering.

Blender Particle Hair Fur Grooming PBR Retopology Character Art Organic Modeling

Technical Features

- Low/high poly modeling with optimized retopology
- UV and PBR texturing with fur detail
- Particle hair and fur grooming in Blender

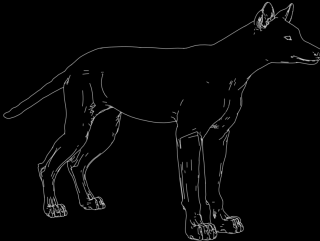
Artistic Achievements

- Organic character modeling focused on anatomy and surface detail
- Particle hair and fur grooming for rendering
- Studio lighting and final presentation

Wireframe & Topology



Wireframe — back view with optimized topology and clean edge flow



Wireframe — three-quarter view showing anatomical structure

Technical Details — Modeling, Texturing & Grooming



Final Render



PBR Detail



Topology



Wireframe

Hard Surface Modeling

Hard Surface Modeling • Blender • PBR • Game Ready

Melika Shahghadami | Technical Artist | 4 Industrial Models



Energy Battery Pack



FUEL Barrel

12K-20K Tris

Game Ready

PBR

Hard Surface



1. Energy Battery Pack



2. Fuel Barrel



3. Industrial Toolbox



4. Industrial Robotic Arm